



Congratulations,

Due to your outstanding academic performance in college, you are eligible to interview for all NUPOC career fields, including a staff engineer position at Naval Reactors (NR) Headquarters in Washington, D.C. I attached a relatively in-depth description of the "NR Engineer" role to this letter. If the job piques your interest, ask your recruiter to set up an informational interview with a Naval Reactors Engineer to learn more about the career field. A member of our team would love to speak with you!

Best Regards,

Payton Alsup
Naval Reactors | Director of Outreach & Interviews
payton.d.alsup.mil@us.navy.mil

Naval Nuclear Propulsion Program Overview

Mission

The mission of the Naval Nuclear Propulsion Program (NNPP), also known as Naval Reactors (NR), is to provide militarily effective nuclear propulsion plants and ensure their safe, reliable, and long-lived operation. This mission requires the combination of fully-trained men and women with ships that excel in speed, endurance, stealth, and independence from logistics supply chains.

NR Headquarters

NR is responsible for all aspects of the NNPP including research, design, construction, testing, operation, and maintenance of the naval reactors aboard Navy ships and ultimate disposition of spent naval reactor cores. This includes responsibility for Department of Energy facilities under NNPP cognizance, regulatory authority for radiological controls involving naval nuclear propulsion, and selection, training, and assignment of personnel. All of this work is accomplished by dedicated research labs, nuclear-capable shipyards, equipment contractors and suppliers, and training facilities, which are centrally controlled by a relatively small NR Headquarters staff (~500 employees) in Washington, D.C.

NR operates under the authority granted by Executive Order 12344 as an integrated organization within the Department of Energy (DOE) and the Department of the Navy (DON). Approximately half of NR Headquarters employees are active duty military personnel, one fourth are DON Federal civilians, and one fourth are DOE Federal civilians.

NR Headquarters Engineers (this could be you!) set standards and specifications for all Program work, while our Field Office representatives monitor work at our prime contractors, prototypes, and shipyards.

To carry out these responsibilities, NR Engineers function as technical managers in a wide variety of disciplines including the following:

- Reactor Design
- Materials Development
- Testing and Quality Control
- Components (pumps, valves, heat exchangers, etc.)
- Electrical Systems / Instrumentation and Control
- Shielding
- Reactor Physics
- Fluid Systems Design
- Chemistry and Radiological Controls
- Electric Power Generation and Distribution
- Embedded Systems and Network Cyber Security
- Facilities Management
- Project / Program Management
- Reactor Safety

A typical Engineer will be responsible for several projects, equipment types or designs. The Engineer has responsibility for technical matters which can entail:

- Reviewing and approving equipment designs
- Allocating project funds
- Technically directing laboratory and contractor efforts
- Ascertaining equipment test requirements and reviewing/approving test results
- Responding to fleet engineering problems by coordinating technical investigations and approving corrective actions
- Determining the scope of work and timelines to support future projects

Based on over seven decades of engineering experience in nuclear propulsion, the NR Headquarters organization exercises exacting control over all aspects of the Program, demanding technical excellence and discipline unparalleled among nuclear programs.

Naval Reactors Field Offices

Naval Reactors has field offices located at its nuclear laboratories, its training facilities, and the public and private shipyards certified to do NNPP work. Field Office representatives serve as the “eyes and ears” of Naval Reactors Headquarters, performing oversight of contractors, private industry, and Navy personnel to enforce compliance with government rules and NNPP standards. NR has over 300 employees at Field Office sites.

Naval Nuclear Laboratory (NNL)

The NNL is a multi-site research, development, and support laboratory for all aspects of naval nuclear propulsion. The NNL is owned by the government and operated by a contractor. Fluor Marine Propulsion (FMP), a wholly owned subsidiary of Fluor, currently holds the contract to staff and operate NNL sites. NR is FMP’s sole customer and controls the work performed by NNL employees. FMP employs over 7,500 personnel at NNL sites.

Bettis Atomic Power Laboratory and *Knolls Atomic Power Laboratory* are the two largest NNL sites by staff size. These NNL laboratory sites develop advanced naval nuclear propulsion technology and provide technical support for the construction, operation, and maintenance of naval nuclear propulsion plants.

NNL assists the Navy in training naval personnel on operational “prototype” reactors at the *Kesselring Site* in West Milton, NY, and aboard *Moored Training Ships* in Charleston, SC.

NNL operates the *Naval Reactors Facility* (NRF) on the Idaho National Laboratory near Idaho Falls, ID. NRF receives, examines, and prepares spent naval nuclear fuel for temporary storage.

Nuclear Component Procurement Organization

The NNPP has a dedicated prime contractor that is responsible for managing the Program’s industrial base. Bechtel Plant Machinery, Inc. (BPMI), a privately owned Bechtel operating

division, currently holds this contract. NR is BPMI's sole customer and controls the work performed by their employees. BPMI has roughly over 1,000 employees.

BPMI is involved in the design, purchase, quality control, and delivery of major propulsion plant components for installation in nuclear-powered submarines, aircraft carriers, and prototype plants. The organization manages contracts with most of the key vendors used to fabricate NNPP equipment. It has operations at two locations: Monroeville, PA, and Schenectady, NY.

NR Engineer Interview and Selection Process

The Navy commissions 15 NR Engineers each year through the NUPOC Program. NR conducts a thorough screening and interview process to select this small cohort of high-potential individuals. Before interviewing, interested applicants must coordinate with their recruiters to:

- Obtain medical clearance to commission in the Navy (known as an "N3M letter")
- Initiate a security investigation
- Submit all application paperwork to Naval Reactors and receive a corresponding Professional Recommendation (PROREC) to interview for the NR Engineer position

Once you meet these requirements, your recruiter will work with the NUPOC Program Manager's Staff to coordinate your interviews. These interviews are conducted in-person at NR Headquarters in Washington, DC. Interviews are typically structured as follows:

- You will have separate technical interviews with three senior NR Engineers. Each interview will last 45 to 60 minutes. Questions typically focus on technical topics you studied in college or graduate school and, at a minimum, cover basic calculus and calculus-based physics topics. In addition to technical topics, these interviewers will probe your motivation to join the Navy, your understanding of the NR Program, and other whole-person topics.
- Based on your performance in initial technical interviews, you may be asked to conduct a fourth technical interview.
- You will have a final interview with the Director of Naval Reactors, a four-star Admiral. This interview will last 5 to 20 minutes and will cover non-technical topics of the Admiral's choosing. He makes the ultimate hiring decision on the spot.

Before you conduct these formal interviews, you'll have the opportunity to conduct an informal, informational interview with an NR Engineer (typically from your alma mater or with the same major). The purpose of the informational interview is to give you a chance to learn more about NR (culture, expectations, job description, etc.) ahead of your formal interviews. Contact your recruiter to schedule an informational interview at any point in your application process.

NR Engineer Commissioning and Training Pipeline

If the Admiral selects you to commission as an NR Engineer, you'll sign official paperwork to affiliate with the NUPOC Program and will begin to earn and accrue benefits (\$15K hiring bonus, monthly pay, healthcare, paid time off, etc.).

If you are still in school at this point, your NUPOC affiliation will boil down to these core requirements: (1) go to school, (2) get good grades, (3) don't do drugs, and (4) don't end up in jail. Your recruiter can cover specific NUPOC requirements with you separately. After graduating, you will commission and begin the following training pipeline.

Officer Development School 5 Weeks

After commissioning you will attend Officer Development School (ODS) in Newport, Rhode Island. This training is not specific to Naval Reactors. It provides all Restricted Line officers (roughly equivalent to non-combat officers) with a basic introduction to Naval Leadership, Naval Administration, Naval Organization, Sea Power, Military Law, Military Indoctrination, Naval Warfare, and Damage Control.

Initial NR Engineer Technical Training ("S6G") 9 weeks

After reporting to Naval Reactors, every Engineer will attend an initial nine-week compilation of technical courses collectively referred to as "S6G." S6G courses cover material ranging from reactor physics to radiological controls and are designed to provide all Engineers a baseline understanding of core Program technical topics before attending Bettis Reactor Engineering School (BRES).

The training regimen includes 18 to 20 hours of classroom instruction per week with assigned homework and exams. S6G is typically held twice a year, with classes beginning in March and September. New Engineers are typically placed into the first class that convenes after they report to NR.

Before S6G starts, Engineers will have a standard engineering workload commensurate with their current level of training. Engineer workloads are tapered during S6G.

Prototype Training 2 weeks

After S6G and prior to BRES, Engineers will visit a prototype at the Kesselring Site in New York or a Moored Training Ship in South Carolina for two weeks. Engineers will have the opportunity to see and operate real nuclear equipment and systems. Training will consist of lessons, demonstrations, and hands-on exercises coordinated by the plant staff with oversight from the local field office.

Bettis Reactor Engineering School (BRES) 6 months

Every Engineer will attend BRES after completing S6G. BRES is taught twice a year at NR Headquarters via VTC, with classes beginning in March and September. Engineers who begin S6G in September will typically begin BRES the following March. Those who begin S6G in March will likely attend BRES that September. Classes are taught by NNL employees and focus on nuclear reactor design, construction, and operations.

Shipyard Training 3 weeks

Four public and two private shipyards service the Navy's nuclear-powered submarines and aircraft carriers. In general, our private shipyards build new submarines and aircraft carriers; our public shipyards maintain them. NR conducts oversight of all of these processes. Immediately following BRES, you will travel to one of our shipyards for three weeks to learn about NR's oversight functions from local Field Office personnel.

Expended Core Facility (ECF) Training 1 week

Immediately following Shipyard Training, Engineers will travel to the Naval Reactors Facility (NRF) in Idaho to observe operations at the ECF. Following this training, Engineers will return to Washington, D.C., where they will fulfill the remainder of their five-year service obligation.

Post-BRES Seminar Series (PBSS) 6 months

Every Engineer will complete the PBSS after returning to Washington, D.C. The series consists of 28 lectures delivered between May and November every year. Each lecture lasts one to two hours and is designed to help Engineers understand the roles and responsibilities of every section at Naval Reactors.

Career Progression

Naval Reactors is a relatively flat organization with very few degrees of separation between the Director and the most junior Engineer. The Headquarters is organized into roughly 20 departments (sections), each with its own department head who reports directly to Admiral Caldwell. Every section is composed of smaller groups depending on the responsibilities of each department, each managed by a Group Head. NR Engineers begin their careers reporting directly to a Group Head.

Every Engineer is assigned to a headquarters section based on NR's staffing needs and the Engineer's major. Engineers will likely remain in their initially assigned section for the duration of their five-year military service obligation. Each section runs its own qualification process to guide new Engineers towards meeting important career milestones.

Engineers promote along the nominal Junior Officer timeline, advancing to Lieutenant Junior Grade (O-2) after two years of commissioned service (YCS) and to Lieutenant (O-3) after four YCS. Associated pay scales are published online. NR Engineers are required to laterally transfer to another officer community or resign at five YCS. There is no path for Engineers to extend much beyond five YCS or to promote to LCDR. Limited lateral transfer options exist (see below) so the majority of Engineers resign their commissions and either continue to work for NR as civilians (in the Department of the Navy (DON) or the Department of Energy (DOE)) or seek employment opportunities elsewhere. Many Engineers continue their Naval service in the reserves.

Engineers who transfer into the DON or DOE are usually hired at the GS-14 level and earn a starting salary of ~\$125K a year before taxes. Once they have completed their initial five-year military service obligation, Engineers are eligible to compete for various promotions and transfers within Naval Reactors Headquarters and its Field Offices.

Lateral Transfer Options

(1) Naval Reactors Engineering Duty Officer

- The Naval Reactors Engineering Duty Officer Community is a special subset of the wider Engineering Duty Officer Community. The Community does not receive manning from direct accessions; it is completely manned by lateral transfers from the 1220 community.
- On average, one to two Engineers laterally transfer into the NR Engineering Duty Officer Community every year. This is roughly 9% of the NR Engineer population that reaches 5 YCS every year and must laterally transfer or resign.
- NR Engineering Duty Officers can promote up to CAPTAIN (O6) and complete a full 30-year career.

(2) Other. NR Engineers have recently had success laterally transferring into the following officer communities:

- Human Resources
- Intelligence
- Information Warfare
- Cryptologic Warfare

Graduate Education Opportunities

Naval Reactors has a generous tuition assistance program, which employees can utilize to pursue any training related to their work (including graduate-level courses).

BRES graduates receive the equivalent of a post-graduate certificate in Nuclear Engineering from the NNPP. The following schools offer post-graduate transfer credits to BRES graduates: Catholic University, Duke University, Naval Postgraduate School, North Carolina State University, Old Dominion University, and Penn State University.

The NR Dress Code

NR Employees are rarely required to wear a uniform. New Engineers work in an office environment where the dress code ranges from business formal to business casual attire.